HANNAH NAGPALL

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SUMMARY

Backend-focused software engineer (GPA 3.92 Honors, TAMUK) with production experience in Java/Spring Boot REST APIs, Python ML pipelines, and spectroscopic data analysis. Co-author of two IEEE-track research papers.

EXPERIENCE

Backend Development Intern · GSPANN Technologies, Inc. *May – Aug 2025*

- Developed backend components for an e-commerce platform using Java, Spring Boot, and MongoDB, including user authentication, product catalogue, and order handling.
- Built and structured RESTful APIs for core workflows such as user login, product retrieval, and order processing.

Research Assistant — Raman Spectroscopy · TAMUK Dept. of Chemical & Natural Gas Engineering *Nov 2025 – Present*

- Built a spectral analysis pipeline for multi-component hydrocarbon mixtures, implementing **peak detection, Lorentzian peak fitting/regeneration, and NNLS-based composition ratio estimation** across 500–3500 cm^{-1}.
- Applied preprocessing techniques (baseline correction, normalization, smoothing) to improve peak resolution and stabilize decomposition of overlapping spectra. Validated results against reference spectra with **<5% composition error** across multiple fuel samples

AI in Retail Technology Extern · The Home Depot (Extern) *May – Jul 2025*

- Benchmarked 3 AI risk frameworks for retail supply chain; delivered Risk-Landscape Summary adopted as reference material across the extern cohort.

VC Research & Pitch Desk Extern · Virtual *Jul – Aug 2024*

- Produced market analysis and a 2-page investment summary for an energy-sector deal, synthesising competitive landscape and valuation benchmarks for investor presentation.

RESEARCH & PUBLICATIONS

Decoupling Transient Instability & Steady-State Persistence in Nonlinear Multi-Agent Influence Networks *IEEE TCNS (Under Review)*

- Built a nonlinear influence–recovery simulation framework to analyze stability dynamics in multi-agent networks across different graph topologies.
- Full reproducible codebase on GitHub; paper open-access: https://zenodo.org/records/19323567

Behavioural Robustness in Temporal Health Risk Prediction under Lifestyle Intervention *Submitted: IEEE JBHI*

- Developed LSTM/Transformer models to evaluate behavioural robustness of temporal disease risk predictions under simulated and real lifestyle interventions using longitudinal health datasets.
- Open-sourced training, evaluation, and dataset-generation code; paper: https://zenodo.org/records/19442886

PROJECTS

Raman Fuel Analysis System · Python · SciPy · pandas · matplotlib *2025 – Present*

- Developed a spectral analysis pipeline for multi-component hydrocarbon mixtures, incorporating **peak detection, Lorentzian peak fitting/regeneration, and NNLS-based composition ratio estimation** across 500–3500 cm^{-1}. Engineered preprocessing and normalization workflows to improve peak resolution and ensure stable decomposition of overlapping spectra

F1 Race Predictor · Python · scikit-learn· pandas *2025*

- Developed a machine learning model to predict Formula 1 race outcomes using historical race data (lap times, qualifying positions, weather, and driver performance). Performed data preprocessing, feature engineering, and model selection (e.g., regression/classification models) to improve prediction accuracy.

EduQuest Advisors Platform · HTML · CSS · GitHub Pages *2025 – Present*

- Founded and shipped a live web platform for India's first à-la-carte college tennis placement service; operates with a 3x/week social content pipeline targeting AITA-ranked players.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, HTML, CSS
Backend: Spring Boot, REST API
Databases: MongoDB, PostgreSQL, MySQL
ML / Data: scikit-learn, pandas, NumPy, SciPy, matplotlib, NNLS
Tools: Git, GitHub

EDUCATION

B.S. Computer Science (Honors) · Texas A&M University–Kingsville *Aug 2025 – Present | GPA: 3.92*
B.S. Computer Science · Alabama A&M University · NCAA D1 Scholar-Athlete (Tennis, Full Scholarship) *Aug 2023 – May 2025*
Awards: Honors College Scholar · NRF Ray Greenly Scholarship · Presidential Freshman Award · Presidential Silver Medallion

LEADERSHIP & COMMUNITY

Community Volunteer · Samidha Welfare Foundation *Ongoing*

- Contributed to community outreach and welfare initiatives at a nonprofit serving underprivileged communities; supported on-ground coordination and event execution across service drives.